

BATCH 18 PRESENTS

ECHOS WITHIN

AUDIO STEGANOGRAPHY USING LSB

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Team Name

PROJECT OVERVIEW

Steganography is the art of hiding secret information inside a non-secret media file. Unlike encryption, it doesn't attract attention.

KEY FEATURES

- > Hides text inside audio using **LSB** (Least Significant Bit).
- > Supports **.mp3** (auto-converted) and **.wav**.
- > Web-based interface built with **Python Flask**.
- > Ensures secure, invisible communication with negligible audio distortion.

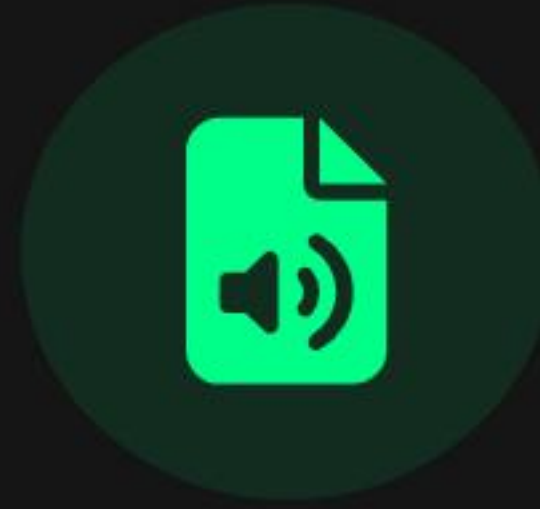


MOTIVATION & NEED



SECURE COMMS

Encryption is visible and raises suspicion. Steganography is invisible, making it ideal for covert communication channels.



DATA HIDING

Audio files are large and complex, allowing for significant amounts of data to be hidden without noticeable quality loss.



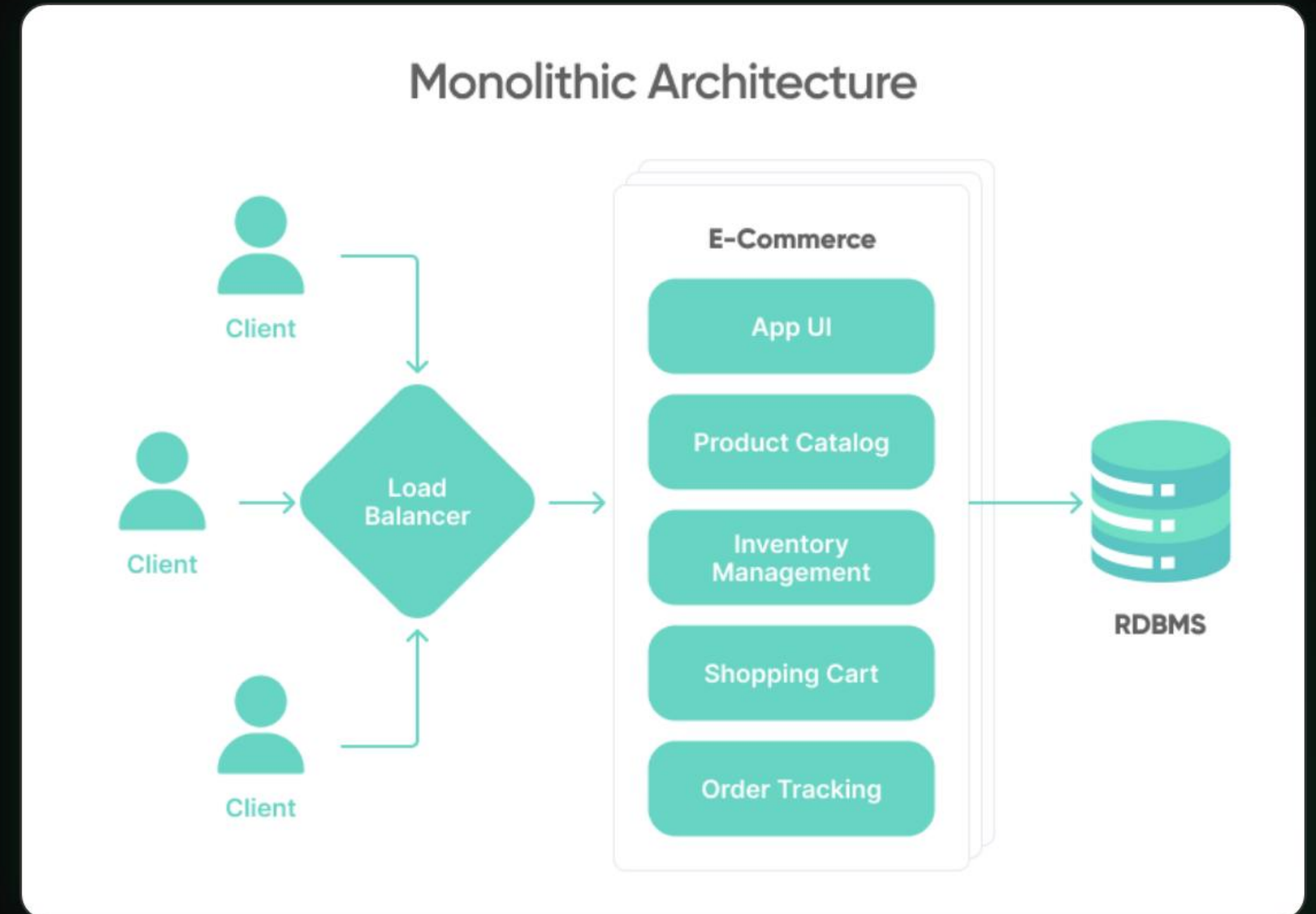
PRIVACY

Enhances cybersecurity protocols by adding a layer of concealment to sensitive information transfers.

SYSTEM WORKFLOW

The application follows a linear, user-friendly process designed for efficiency.

- > **Input:** User uploads Audio + Secret Message.
- > **Processing:** MP3 converted to WAV if needed. LSB engine encodes bits.
- > **Output:** Encoded Audio file generated for download.
- > **Decoding:** User uploads encoded file -> System extracts and displays text.



THE LOGIC: LSB ALGORITHM

ENCODING

- > Convert secret message to `binary bits`.
- > Read audio frames from the WAV file.
- > Replace the **Least Significant Bit** of each audio byte with a message bit.
- > Append delimiter `###END###` to signal stop.

DECODING

- > Read the encoded audio file frame by frame.
- > Extract the LSB from each byte.
- > Reassemble bits into characters.
- > Stop reading when `###END###` is detected to reveal the message.

TEAM CONTRIBUTIONS

MEMBER	ROLE & CONTRIBUTION
Savyam Shukla (Lead)	Core LSB logic, Bit manipulation, Encoding/Decoding Engine.
Aryan Chauhan	MP3 to WAV conversion (pydub + FFmpeg setup).
Harsh	Frontend routing, Template linking, Navigation structure.
Zaman Ahmed	Backend '/encode' route, File upload handling.
Shaurya Pathak	Backend '/decode' route, Error handling, Message display.

IMPACT & FUTURE

Current Applications:

- > Digital Watermarking for Copyright.
- > Secure Military/Intelligence Comms.
- > Data Integrity Verification.

Future Scope:

- > Implement **AES Encryption** before hiding.
- > Support for hiding Image files inside Audio.
- > Development of a mobile application.



Q & A

Thank you for listening.

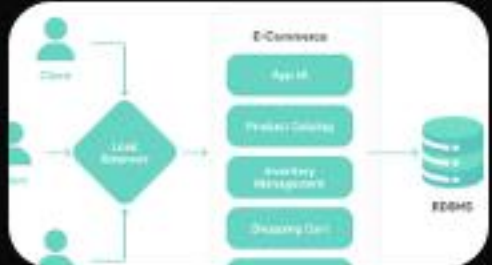
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IMAGE SOURCES



https://static.vecteezy.com/system/resources/previews/014/675/666/large_2x/abstract-colorful-rhythmic-digital-sound-wave-with-binary-code-on-black-background-sound-waveform-digital-information-photo.jpg

Source: www.vecteezy.com



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